

DSA 8405 Probability and Stochastic Processes- 20250918_170514UTC-Meeting Recording

September 18, 2025, 2:29PM

2h 35m 22s



Odhiambo, Collins 1:57

So good evening everyone.

I think we're going to start again our class, so let's give the rest like.

Few minutes, like 4 minutes so that we start at 5, we give them like we start at 5:35.

All right. So I think we're going to move on. Um.

I want us today to do just a recap of what we covered last time. I realized that probably there was some pieces here and there that I need to make clarification.

So let's just do a recap of Uh.

Uh, what?

What we covered last time?

OK, so we looked at the steady state. What we see that our system is in steady states if so you have you have people coming in or entities entering a system.

Uh, so that system at at at state I and then.

Within the same period of time you have exits, so you have entries at at the rate of λ and then at the same time you have exits at the rate of μ .

Then.

So this is when you consider the, let's say the.

The one hour period.

For example, in the bank within the first one hour when the bank opens its doors for customers, within the first hour you'll have customers coming in and then at the same hour you'll have some customers leaving.

And then if.

The following hours. So probably this one. The business open at 9:00 AM.

To 10:00 AM. OK, so they are now if you look at the next one hour here between 10:00 AM.

To 11:00 AM.

You still also will have some customers coming in. Remember there was people inside and then you also have some customers going out.

OK, so this is like a very typical bank. Then another one hour after that, let's say that is state three you.

Customers coming in μ and then you have customers going out.

So the rate of exiting is μ , the rate of coming in is λ .

So that that will be rate per minute or per second. How many customers are coming in? The rate at which they are coming in is λ .

The rate at which they are going out is μ .

OK, so we say that this system is in. So we consider what we call, we consider what we call steady state.

So we consider what we call a steady state. So this system is in steady state if it's in steady state if the rate of coming in.

μ .

Is the same as the rate of.

Going up.

So if the rate at which customers come into the system is the same as rate at which customers go out of the system.

The system is so that's what we consider and we are assuming this is an MM one sort of model.

So we are considering this is an M1 queuing theory model. OK, so if this is the case then we can make the subject. So that's what we were doing last week. So we can make the subject so PM the subject. So if you make PM the subject.

Uh, you have.

You have these.

So if you make PN the subject, you have that.

Right. Remember PM is a this is a probability that there are N customers or can NND. Is in the system.

A PN is a property that there are N entities in the system. Now when we make PN the subject here, then we end up with what we call a difference equation.

So we end up with what we call a difference equation.

OK.

Uh, so we end up with what we call a difference. So to solve a difference equation we solve it using iterative methods.

Iterative uh method.

OK, so to solve the difference equation we use iterative method. In iterative method is very simple. You assume N is 1.

So if you assume N is 1, you end up with P_1 is equals to λ over μ .

So if you assume N is 1, that's what you get. If you assume N is 2, you end up with P_2

is equals λ over μ P1 one.

OK, but but P1 is already given. P1 is already given.

So you just replace. Therefore P2 becomes λ over μ .

I'm now on a new P0.

So that's what we were solving last week, which eventually this if you open up that bracket, then you have.

Uh, you have that.

OK, so if you do the same for P3, you will have the same.

OK, you'll end up getting that because now P3 will give you P3 will give you P is yeah, λ over μ P2, but P2 is already known OK.

So, so P3 will give you that.

And therefore PN.

Example generalized, so we can generalize PN and NP0. So that's what we showed last time. And of course we went further to explain that if that's the case.

Then we know the sum of all probabilities. OK, from the principal probabilities that if you sum all probabilities you should get one. So then if PN is that, that means then if we sum.

If we sum N from zero to Infinity, PN is the same as you sum. So if you introduce a summation.

Oh.

NP 0 We should give you one.

It so that should give you one.

We went further to work on that. So if you bring P0 outside and you sum N is equals to zero to Infinity of.

Of that that expression you should get one. OK and because this is a queue, since this is a queue, a queue means that the rate at which service is done. Remember this is the rate.

At which customers arrive, right? So this is the arrival rate.

And this is the service service today of service mechanism.

Right, so the new is a service mechanism. Uh, λ is a rival rate.

Right. So for that to be a queue, it means that this is more so for that for queue to exists.

OK for Q to to to exists.

It means that this is greater than the rate at which customers arrive. So it takes long time to serve our customers compared to the rate at which the customers can arrive.

And if that's the case then.

Uh, then λ / μ is a fraction.

So is a .

OK, so that means then if you unpack this summary here, if you try to unpack this component here, you will end up with. So you put where there's N you will start with 0 so you will have one.

Plus λ / μ plus λ / μ^2 .

So you'll go that way up to some N λ / μ^N .

So this is these are geometric progression.

That's a geometric progression and there's some of our geometric progression of our GP.

So if if if a GP is made-up of $1 + R$ where R is the common ratio.

So if you if if you get the sum of a GP, assuming that R is a fraction, R is a fraction, it's R lies between one and zero. OK, so if you sum up.

A geometric progression. Assuming R is a fraction, you should get R plus that something like this.

And this one you did in your elementary. You must have done this in high school or I don't know your first basic mathematics, algebra or geometry.

OK, so if if you saw my geometric progression you get this, but if that N .

Approaches Infinity.

So if that N approaches Infinity, that means you're powering a fraction. This part here you're powering. If you power a fraction to a very big number, it tends to 0. So as N tends to Infinity here.

The whole sum of our geometric progression is just $1 - 1$ over.

OK, so if you sum a GP as long as N is greater than tends to Infinity, you have that and so then P .

Uh.

E0.

SK **Stephen Korir** 19:41

Hey, Prof, I think we've lost you.

Or is it on my end only?

The.

OC **Odhiambo, Collins** 20:08

Sorry guys, can you can you hear me now? I think I got, I got dropped.
I got dropped.

SK **Stephen Korir** 20:16

Yes, yes, we can now hear you, Prof.

OC **Odhiambo, Collins** 20:19

OK, OK. Sorry guys, I think I just got dropped by the system.

OK.

So.

All right, so.

Some this this will give you so this will give you like a GP. So if you unpack this if you unpack this sum that that expression.

So where there's N you put a zero, so we'll start with that plus I plus pi over.

OK.

Squared that way to Infinity, OK.

And this.

It should be all he said.

Now you should give us more about wireless instead of.

So this is should um.

Give you one.

Making P the subject P0 the subjects, then you'll have one minus.

So that.

So that's your PN.

Love over me and.

Instead of P0 we have.

So that now becomes your your RP. So remember PM is the probability.

That there are N customers, N entities in the system. So looking at a bank call, that's a probability that we have N customers in the system.

So to solve this problem I.

Sure, yeah.

Like um.

Um, question one. Let me just give an example, yeah.

Customers arrive.

In a bank.

To the rates.

So from previous.

So from previous previous data it shows that every one hour we have 12 customers arriving.

It takes.

It takes, let's say 15. Let me just say per minute. Let me work with minutes per minute with this network. So it's within per minute.

It takes a lot of customers to write them. It takes 15 minutes.

Uh, self.

8.

On average.

BO **Bernard Ogol** 24:36

Excuse.

Excuse, I think we can just hear what you're saying, but then what you're presenting is not.

OC **Odhiambo, Collins** 24:43

Yes, yes.

BO **Bernard Ogol** 24:45

Yeah, you can only hear what you're saying, but then what you're writing you can't see.

OC **Odhiambo, Collins** 24:52

Alright, let me.

Maybe share again.

Can you see it now?

BO **Bernard Ogol** 25:07

Yeah, yeah, yeah.

OC **Odhiambo, Collins** 25:10

OK, so I'm doing, I'm just creating a question and trying to see if we can solve this question. So customers arrive in a bank at the rate of 12, yes.

SA **Straton Amodora** 25:21

Hello.

I think you've gone ahead. It'll be better if you first call back because the screen was stuck. Michelle was stuck.

OC **Odhiambo, Collins** 25:35

What's up there? The the whiteboard? You are not seeing the whiteboard.

SA **Straton Amodora** 25:41

No, it it was it was initially stuck. So there's some others that we are writing.

Yeah, there is OK.

OC **Odhiambo, Collins** 25:52

OK.

All right, so I don't know why today's looks to be misbehaving.

Probably there's a there are a lot of students around.

SA **Straton Amodora** 26:11

A bit, a bit down.

No, not down. Absolutely.

OC **Odhiambo, Collins** 26:23

Oh, oh, you want to see, yeah.

SA **Straton Amodora** 26:26

Yeah, that that day's OK.

OC **Odhiambo, Collins** 26:29

OK, OK. So we were just trying to demonstrate what we did last week by doing a recap. So, so we ended up up here we started with.

Steady state, then we were able to show that if you consider steady state, you'll end up with a difference equation and then in this difference equation to solve it, you can solve it directly and when you solve it directly, you'll end up with.

With this expression, yeah, OK, so then working for P not, finding P not, we use

geometric progression, then you'll end up finding that your expression is eventually the. So this becomes.

The probability that there are N entities in the system.

So I'm just trying to give a practical example on how you can solve practical equation in a queuing system. So customers arrive in a bank at the rate of 12 per minute. So we have 12 customers arriving per minute.

It takes an average 15 minutes to serve each customer. OK, so if we assume MM one. In one.

Model.

Or queue. We assume an M1 queue compute.

Uh, I follow.

The the number of customers, the probability that they have the probability.

But they are our five customers.

In the system.

SK **Stephen Korir** 28:43

Hey, Prof.

OC **Odhiambo, Collins** 28:47

Yes.

SK **Stephen Korir** 28:48

Are you Are you writing in the whiteboard?

OC **Odhiambo, Collins** 28:53

Yes, yes, you can't see.

SK **Stephen Korir** 28:56

On my side I cannot see. I think it's frozen. I'm not sure about my colleagues.

OC **Odhiambo, Collins** 29:05

Oi.

SA **Straton Amodora** 29:18

Also, my side is fixed, not moving.

OC **Odhiambo, Collins** 29:20

Where is Riya again?

Is it OK now or still?

Uh.

SK **Stephen Korir** 29:52

I think your voice too is equally patchy.

OC **Odhiambo, Collins** 30:01

OK.

SK **Stephen Korir** 30:02

Maybe it's an Internet connection issue.

OC **Odhiambo, Collins** 30:03

OK, so.

Me.

Let me my Internet.

Alright, so this should be my.

And now on again.

No.

Now OK.

JM **Jackline Mboya** 35:48

Good.

OC **Odhiambo, Collins** 35:51

You can't see the.

Can you see the whiteboard?

SA **Straton Amodora** 35:56

I can just see a blank, but not a white one.

KR **Kennedy Rutto** 35:58
No, not yet, not yet.

OC **Odhiambo, Collins** 36:02
Oh.
What's happening here?
Still not seen?

SA **Straton Amodora** 36:28
Still not visible.

OC **Odhiambo, Collins** 36:30
Oh, OK, but if I write something like uh.
EN can you see something that I'm writing?

SA **Straton Amodora** 36:43
OK, we can't see anything. You can just see a white space, but not a whiteboard.

OC **Odhiambo, Collins** 36:49
Hello.
Well.
OK.
All right, so I don't know what's happening today.
I need to check with the ICT guys here later.
I don't know what's happening today, but I wanted us to finish up that section. Um.
The whiteboard is still.

CO **Collins Odhiambo** 47:21
Trying to use rather than use the app. I brought the guys from ICN like they're not.
They're saying there's nothing wrong. So I'm hoping this will work now. Can you guys hear me?

KR **Kennedy Rutto** 47:39
Yes, we can.

co **Collins Odhiambo** 47:42

OK, uh.

I'm using the browser instead of the.

Instead of the app.

There's something that uh.

Can you see my screen now?

KR **Kennedy Rutto** 48:29

Yes.

co **Collins Odhiambo** 48:31

OK, so I I was, I I'd written an example. There are customers arrive at a bank at the rate of 12.

And then this should be.

It should be N here.

1 minus Lambda over MU as the formula. So can you see like the formula I'm writing that all the writings that are there like the example a customer. Can you see those right?

KR **Kennedy Rutto** 49:04

Yes, yes.

co **Collins Odhiambo** 49:05

OK. OK. Thanks.

OK, so so the first question is find the probability that there are 5 customers in the system find.

Uh, the probability.

That the is either.

Uh, find the probability.

That's they are are at. They are at most.

For customers in the system.

Just one, uh, find a ability.

That's there.

OK, so find a property that there are at least.

Assumes.

In the system.

I'm honey, is it fine now?

Are you OK or am I? I just wanted to verify. OK, OK, OK, so we.

SA **Straton Amodora** 51:19

It's OK.

OC **Odhiambo, Collins** 51:20

OK.

CO **Collins Odhiambo** 51:25

And here that we want to solve. So the first question is to find the property. There are five customers in the system, so.

What we need is we need we need our.

In the same unit. So if arrival is in minutes and service is in hours, you have to convert so that you have the same units. So this time we have arrival in minutes. Service is done in minutes.

So the service is done in minute and arrival is in minute. Then we are good to go as long as we have λ and we have n now to solve the first part of a.

So we need P_5 probably that there are 5 customers in the system. P_5 can be found by by by that.

It.

One minutes, 12/15.

OK, so if I try to, I try to.

Reduce this to work on this. This is the is it three or five? Five. It's 4 should be 4.

It should be 4, not 5.

When you need to on your flight.

So you can work on and plug in and get that. So that's how you get the probability that there are 5 customers in the system. So you just use the formula and you get the answer. So if you plug in here you can get.

The answer for question A, so question A, the answer is $4 / 5^5$. So you can work on that and then multiply by by that by just replacing it into the home.

Because the formula is $P_N = \frac{\lambda^n}{\mu^n (1 - \lambda/\mu)}$.

As a formula, so you just need to work on that to get the value. This is for part A part

A. OK so for Part B.

Probability that they are the server is idle. The probability that the server is idle. That means there's nobody in the system. So the probability that the server is idle means that there's nobody because the server is only idle and there's nobody inside the system.

So that is the probability P_0 .

When there's nobody in the system, we're looking at the probability that there's zero entities in the system.

0 is $1 - \lambda / \mu$ which is $1 - 12 / 15$ which is $1 / 4$.

So that's the probability that the server is idle at any time.

Right. And then the third bit here is find the probability that they are at least or at most they are at most 4 customers. At most 4 means what means 0.

At least 0, 1, 2, 3 and four. So at most means that either.

At most 4.

That means so we either have there's zero or there's one or there's there are two customers or there are three or there are four, so.

What's the meaning of at most? At most means it's either zero or one or two or three or four.

OK, so so for P_0 we have $1 / 4$ plus P_1 we have $12 / 15$.

$1 / 4$ plus P_2 we have $12 / 15$ two $1 / 4$ plus P_3 we have $12 / 15$.

15 three $1 / 4$ plus P_4 is $12 / 15$.

Call.

OK, so you need to plug this in and get the value at at most 4 means $P_0 + P_1 + P_2 + P_3 + P_4$.

OK. So that's that's for part C for Part D, the probability that they are at least at least five.

You can do it on the reverse at least five means 567 going on to Infinity.

OK, so at least five.

Um, at least five. So that's part D.

At least five means so you can do the reverse, so you do $1 -$ minus.

They reverse, OK.

It's at least five means 5678 up to Infinity, so you can do just one minus the one that is not included in that.

So basically what the answer you get here after you compute this part here.

This this all part C that answer you get is what you bring so that you do one that

answer that you got in C.

Right. Yeah. So that's for probability. So that was one part that I wanted us to go through. The other bit is now for the for the means average entities in the system, average entities in the.

Queue and then the the Little's law which tells us the average waiting time in the system and the average waiting time in the queue. So the other bit was the average entities in the system.

The average entities in the system we use the same concept of mathematical expectation. So the average entities in the system we use the same concept of mathematical expectation.

And then.

We already know what is our ρ , so we already know what is our P_N . So our P_N is N is λ . Our P_N is.



1:02:45

Was to summation.



Elias Shamalla 1:02:49

As we can see.



Collins Odhiambo 1:02:50

OK.

All right. So just a recap of what we were doing last week. Again, if you work on this, if you work on this equation here, that means that you bring.

The constant outside, which you bring the constant outside so that you have N is equals to zero to Infinity summing.

Something that OK, so we we found out that if you open this summation here, you end up with what we call an increasing, increasing geometric.

Geometric.

progression.

So increasing Jamaican tick programming.

So if you if you unpack this, if you unpack this.

Please.

This expression here you'll end up with an increasing and we saw that the way it will be, it will be.

So if you put 00, you have 00, so you have 0 plus if you put 111 N where N is 1, so you have that.

Uh, so if you put N is 2, so you have two.

But.

If you put any three, so you have three that

After it to Infinity. So this one is the same as if you try to if you try to open it.

Let me give you that last two.

And over me.

OK.

Yeah.

I'm not on you. Yeah, so doing things, so you'll end up with.

This part here geometric is an increasing geometric probabation and the sum sum of an increasing.

OGP.

It's just one over 1 minus ratio than you square.

So it's $1 - 1 / 1$ minus the common ratio. Then you square it. That's the sum when increasing progression.

Uh, so so then.

If that's the case.

If we replace this expression up here, then we'll have east of north.

Um, has this mean as that?

Plus this, so you'll have that and then here you'll have.

They're one minute.

Mhm.

Right, which?

$1 / 1$ S lambda over R and then that one if you do that and you bring that.

Yeah, then you do, then you do that, you bring back. So this this.

This this this will eventually give you so if you the same is more the rest is more of.

So you bring that here, you bring that here.

Let me see. So we have that out, that out of that list.

Yeah, so it should should give us that's the.

So that's the mathematical expectation, or rather the average time in the system. So if you do the same for waiting time in the queue, the E.

And one for the queue now, because for the for the queue, you don't consider the password as being the as being the queue.

So you have subtract one because one person is being served. So whoever is being served is not considered to be part of the queue. So when you're doing for the queue, so you just do the same, the same mathematical expression here. So we do. And is equals to 0. Now we start from one, not really zero because you cannot have negative person and then you have P and.

Which which which eventually gives you N is equals to $1 - N^{-1}$.

So this part was N one over Λ one.

So that part was that. Again, if you bring out the constant, so you'll have. So if you bring out the constant, then you have that and then you work on whatever.

You want to work on. So there are different ways. Sorry, there's there are different ways of.

Looking at this, you can either work it that way or you can have general reasoning where you consider one person who is being served is not is not being one person who is being served is not in the queue. The other way would be if you open up this bracket. There are different ways of killing the cat, yeah.

You can still open this bracket here and then you have N is equals to $1 - N$ here P_N minus you sum N is equals to 1.

P_N So if you open this bracket here, you can end up with that and then this part here.

Because this is the sum from one to Infinity is the same as you do the the opposite.

$1 - P$ I think this is 1.

Awesome.

So you do. Yeah, you can. It's the same as $1 - P$ zero.

OK. And then this part now is the is like the mean, but it's like what we did for the mean is only that now we are starting from one instead of 0.

And you find out that if you do.

Let me test your. Let me test your algebra.

Is is this the same as?

This.

Are these two the same?

Let me test your algebra, algebra.

Are we still the same?

We're not. Are those two the same?

BO **Bernard Ogol** 1:11:38

No, they're not this.

CO **Collins Odhiambo** 1:11:40

Why?

Why?

I need something from one. Tell me why the other.

Anyone else?

SA **Straton Amodora** 1:12:21

I think that they seem.

CO **Collins Odhiambo** 1:12:26

Yes, they are actually the same.

Plus $2P_2$ plus $3P_3$ that we're going that way forever. OK, then try this one. Start with 0, so we'll have $0P_0$ plus $1P_1$.

Plus $2P_2$ plus that way forever at least to the same.

They're the same because this zero times this will cancel.

So you'll end up having in the scene.

So can you see that there? They are actually the same.

BO **Bernard Ogol** 1:13:20

I was in it. I was in it.

CO **Collins Odhiambo** 1:13:23

OK, they're actually the same. Where are we? Where are we gone?

BO **Bernard Ogol** 1:13:24

Oh.

CO **Collins Odhiambo** 1:13:32

OK, so they're they're actually the same and that that means that this part here.

Is the same as what we got here.

OK. Because what we've got there, yeah, what we got, yes.

BO Bernard Ogol 1:13:48

So.

Do I may ask?

Can you call it the law of cancellation?

CO Collins Odhiambo 1:13:58

It's just general reasoning, or I would call it general reasoning.

It's like you're starting from one. It's just general reasoning. You're starting from one and when you start from zero because the zero will cancel the P_0 and you start from zero.

But if stuff on one, so you're still getting the same thing.

So so that um, this part here is just just this.

And then this other part here is $1 - P_0$ was $1 - \lambda$ on μ .

FO Fridah Obuchela 1:15:08

Hello, Prof, can you hear me?

CO Collins Odhiambo 1:15:10

Yes.

FO Fridah Obuchela 1:15:12

I think there's an error on the solution for EN.

CO Collins Odhiambo 1:15:21

Oh, it should be what? It should be.

FO Fridah Obuchela 1:15:23

It should be λ over μ minus λ .

CO Collins Odhiambo 1:15:29

λ over μ or EN here will be λ over μ minus. OK, so because because this is a product, let me just finish it here. So this means then that you have.

That then there's new on top. Yeah, so this side should be. Thanks. I think it should be.

Should be Λ over me.

Oh, it should be this way. That's that's what you're saying, which is fine. Yeah, it should be Λ over.

Thanks.

Actually.

Yeah. So yeah, if you put the algebra, this, this meal will be here, then the other one will go on top. So that will cancel with that. Yeah, yeah, it should be this.

It's.

It so that this part here is just that.

That's bad. Then that is stopped.

So that this this will cancel one will cancel there that will cancel. So it will cancel and that will go. So it's so this will generally be that.

Minus the μ is busy.

No, this is.

So that now I have.

μ minus $\mu \Lambda$.

And then Λ squared.

So the answer should actually be what I was getting there. I was, I think I was mixing it up.

Yeah.

Yeah, yeah. So these are the most critical.

Um, you must call you.

Formulas. So this is the average entities in the queue.

OK, and this is the average entities in the system.

So that's the average entity in the system and this is the average entities in the queue and then.

Yeah. So that's the average entities in the queue. And then there is the concept of what we call the Little's law.

Yes, there's there's a concept of the Little's law which says that the average and waiting time or waiting time in the system is basically the waiting time in the system. Is basically average entities in the system divided by the arrival. This one we usually not denote it as L_Q , which is the average entity in the queue and then.

The average entity in the system we denote it as L_S , the length of the system L_S at least the length of the queue and queue. So the Little's laws says that if you want to get the.

Waiting time in the system.

Waiting time.

In this system.

Is just LS divided by Lambda.

Which means then that we can take. We already know our LS, so we could take.

So we could take LLS is just given us this.

Bye.

By that so that.

We have one.

So this is the waiting time in the system.

And then if you want to look at the waiting time in the queue, it's just LQ all over.

All of that.

That which is LQ squared over mean.

Well, that's so, so it will become.

Become that.

OK.

All right, so let's take a short break. Then we come. We do 2 examples, just two examples on how to apply this.

So let's take a short break, like 10 minutes, then we'll come in and do some 2 examples.

All right, so let's give some examples here.

Let's not do some two examples yet, and then I think I'll move on from that.

That should be among the last bits of queuing system.

For us at this level, as I always told you earlier that.

You can always broaden some of these concepts, but for us we touch where we can touch and then we move on. So let's look at some two examples. So let's look at an example here.

Um.

So in this example entities.

Arrive.

And uh, system.

At the rate.

At the rate of.

At the rate of.

Um.

Uh, 6 entities.

Uh, second.

Um.

It takes.

Takes one minute.

To serve an entity.

Assuming.

MM1 assumption.

Find.

One probability.

Ability.

Uh, the Saba.

Is I do?

Um.

Probability.

The server is busy.

B.

Uh, waiting time.

In the system.

Um 4.

Average.

Entities.

In the queue.

5 waiting time.

In the.

Yeah.

Can you confirm? Are you able to see my white board with that question example?

Just confirm 'cause I don't want to. I just want to see if I'm lost or right, I've found.

Can someone confirm?

BO Bernard Ogol 1:36:58

Yeah, we can see it.

CO Collins Odhiambo 1:36:59

OK, OK, so want to solve.

Those five questions. So this is the first thing is to establish what's our meal and so our meal is arrival is 6.

Yeah.

So if our arrival is 6, then what is our waiting time? Waiting time? It's one per minute, so one minute to serve. So you have to convert them to all to seconds. That's like 60 seconds.

Right. They have to make sure they all speak the same language. If arrival is in terms of seconds, then the.

Service time is also must be in in terms of seconds. So once we've established what is our Lambda and what is our new, then we can go ahead and quickly get the answer. So the server is idle probability, so that's is P_0 .

That's like $1 - \lambda / \mu$ and that is just $1 - 1 / 10$, which is.

Is it $9/9$ over 10? The summer is busy. It's just Lambda over meal, which is $6 / 60$, which is $1 / 10$.

Waiting time in the system. That's W_S . W_S is just one of a million, which is $1 / 60$. When the suits.

Average entities in the queue. Average entities in the queue is that. So that's would be $6 / 60 - 6/6$.

Of our 54, that's $1 / 9$ orange entities are waiting time in the queue.

Uh, that's.

That's that.

So that's that's like, um, 6.

OK, very simple, very easy. As long as we we have already established our London and we have established our meeting. So the rest are quite simple to work out, just plugging into the formulas.

And then you to answer, you plug in into the formulas and you get your your answer.

OK, very easy, very simple. What a simple queue. Let me look at second question. Let me check the question. Question 2 now example 2.

OK, so uh, so example 2 customers.

I like.

And the system.

The the right job.

Now 10.

Um minutes.

Like to text.

Uh, so it takes.
12 minutes.
To to serve a client.
So if you assume.

SK **Stephen Korir** 1:41:46
Hi, Chris.

CO **Collins Odhiambo** 1:41:46
Yes.

SK **Stephen Korir** 1:41:49
I think I'm a bit, I'm a little bit lost. Would you go back up to the example that we are looking at? I just want to ask something there.

CO **Collins Odhiambo** 1:42:01
OK.

SK **Stephen Korir** 1:42:01
On the MU and the Lambda.

CO **Collins Odhiambo** 1:42:04
OK.

SK **Stephen Korir** 1:42:06
Yeah.
Couple bits, yes. So we say that when the entities are arriving at a system at the rate of 6 entities per second, that is a Λ , then the MU, then the MU.

CO **Collins Odhiambo** 1:42:15
Mhm.
Mhm.

SK **Stephen Korir** 1:42:25
Will be so say it takes one minute to serve an entity.

co **Collins Odhiambo** 1:42:26

Mhm.

Yes.

SK **Stephen Korir** 1:42:33

So what will be the meal here? Is it 60 or $1 / 60$?

co **Collins Odhiambo** 1:42:38

It will be 60, so you convert that into seconds. So one minute is 60 seconds.

SK **Stephen Korir** 1:42:48

Yes. So if if we convert that to seconds and you say that it takes 60 seconds to serve one entity, so the meal is supposed to be the one entity divided by the the second so that we say.

co **Collins Odhiambo** 1:42:54

Yes.

Yes, yes.

SK **Stephen Korir** 1:43:05

$1 / 60$ entities per second. I don't know if I'm mixing myself.

co **Collins Odhiambo** 1:43:13

No, no, it's you should just take it directly this way, but I get what you're saying. Um. So typically the service mechanism will be the service mechanism will follow what we call an exponential distribution. So because it will follow an exponential distribution. The mean of an exponential distribution. The mean of an exponential distribution is usually one over meal.

SK **Stephen Korir** 1:43:48

Mhm.

co **Collins Odhiambo** 1:43:48

So, So what? What the way you are reasoning is perfect is OK because the mean of

exponential is actually one of a μ . So typically if we look at if we model the service mechanism, it should be $1 / 60$ but.

SK **Stephen Korir** 1:44:03

Mhm.

CO **Collins Odhiambo** 1:44:05

You see, we've already taken care of those. We've already taken care of those in these formulations here. So if you if you come back here in our formulation, if we just try take you back to the top.

SK **Stephen Korir** 1:44:09

Of the.

Mhm.

CO **Collins Odhiambo** 1:44:22

So if we come back here in our formulation, when we consider what we call a steady state and we generalize these numbers, we've already taken care of the exponential bit so that at the end of it we we have the formulas in this form.

SK **Stephen Korir** 1:44:29

Yes.

CO **Collins Odhiambo** 1:44:39

And the formulas. So these formulas that we are seeing here like this one.

SK **Stephen Korir** 1:44:45

Mhm.

CO **Collins Odhiambo** 1:44:45

The the the the reasoning is that μ is an exponential. That's why you see it's coming to one of λ over μ . The μ is an exponential.

SK **Stephen Korir** 1:44:57

OK, so we have the points.

co **Collins Odhiambo** 1:44:59

Yes, yes. So that's a good.

Yes, that has already been taken care of, so you just plug it in, yeah.

Yeah.

Yeah.

SK **Stephen Korir** 1:45:18

Right into the system, 6 entities per second. Then it takes one minute to solve one entity, right?

co **Collins Odhiambo** 1:45:20

Hmm.

Mm-hmm.

SK **Stephen Korir** 1:45:29

I'm I'm also trying to wrap my head around that and see is the system not going to blow up like because every second we're having six entities that are coming in. It's like an inflow of 6.

Entities.

Into the system, but the service mechanism, as you've said, takes one minute to serve an entity. So for the six entities it will take like 6 minutes.

co **Collins Odhiambo** 1:45:55

Yes.

Yes, yes, yes. So that's.

That's true. So it is hypothetical the way we are looking at it because we are assuming an MM one. So in in reality you will have to adjust some some issues like for example here we are assuming it's only one server.

So in reality you may feel there will be more than if you get another server and another server, it may not take 6 seconds, 160 seconds to serve one because there will be multiple. There will be multiple probably people to serve who are serving.

SK **Stephen Korir** 1:46:22

Mhm.

OK.

co **Collins Odhiambo** 1:46:39

So and again we may introduce some components where we have, there's some time where when you have a peak time and there's sometimes when you don't have peak time, it's slower. So if you are to consider all that then.

We will have to adjust our model, so our model assumes.

SK **Stephen Korir** 1:46:58

Mhm.

co **Collins Odhiambo** 1:47:01

That there's only one server.

And the rate at which that server operates, it's it's that the duration it takes to serve one client is longer compared to the rate at which they are coming.

SK **Stephen Korir** 1:47:06

Mhm.

Mhm.

co **Collins Odhiambo** 1:47:18

So that there is a queue and then once they come they form a queue. So those are the assumptions that we call MM1 assumptions. So we assume those type level of assumptions.

SK **Stephen Korir** 1:47:19

OK.

OK, thanks bro.

co **Collins Odhiambo** 1:47:32

But I get, I get, I get you, I get you. Because if you do modelling, usually we start with the most, the most stringent assumption, as in some very stringent assumptions, but. As you as you now come into reality, you begin to adjust the assumptions so that it reflects the real the reality.

SK **Stephen Korir** 1:47:52

Mhm.

CO **Collins Odhiambo** 1:47:57

Yeah. So, yeah, so that's the idea. This, this is just a bit like this is a very simple one where we we we are assuming that those strict assumptions apply.

SK **Stephen Korir** 1:47:58

OK. OK. Thank you.

OK.

CO **Collins Odhiambo** 1:48:11

Yeah.

OK, so um.

We go here.

So I have customers arrive at a system at the rate of 10 per minute, 10 customers. It takes 12 minutes to serve each customer. So if you assume MMM one, then some few questions here.

Uh, so get uh find a find uh so can find.

Um, probability.

This service is busy.

We want the waiting time in the system, waiting time in the queue, no length of the system, length of the queue and then waiting time in the system and then waiting time in the.

Q.

The probably the summer is busy. This is usually what we call the the the intensity of the queue which is just so first of all we need to get Lambda then and the service mechanisms 12.

So this is just 10 or 12.

Which is $5 / 6$.

LS is the length of the system. The average entities in the system is just is giving us this.

So we have 10, 12,

Oh.

10.

Then.

That's 5 LQ is.

That's.

Out drop.

That should be that should be 24. They can make it rather than that. Then they will have waiting time in this system.

And.

Waiting time in the system is just the length over that, which is one of and start.

Is one about two hours minutes then.

102 and then waiting time in the queue.

Should be LQ that, so that's like.

That's like this what you got here 24 divide by 10 divide by or just do it.

Oh, that's that's just that.

Yeah, so that's.

So that's how you work it out and I think we can.

Stop there for the queues, so we move to another stochastic.

Process system.

JM **Joseph Mugambi** 1:52:21

I have a question.

CO **Collins Odhiambo** 1:52:21

So for the Q and yes, yes.

JM **Joseph Mugambi** 1:52:25

In in that other example above.

To what to what my colleague, my colleague was asking is in the Lambda. This one isn't Lambda.

360.

If if, for example, if you're saying it takes 6 entities per second, if I if I was put this per minute, then it means in one minute the arrival rate will be 360.

CO **Collins Odhiambo** 1:52:52

Mhm.

JM **Joseph Mugambi** 1:53:01

Whereas we are serving only one entity per minute.

CO **Collins Odhiambo** 1:53:14

Or so there are 6 entities per second, so six people come per second.

But now, so you want us if we convert that per minute, that's what you're saying.

What?

JM **Joseph Mugambi** 1:53:31

Yes.

CO **Collins Odhiambo** 1:53:35

So if you convert that per minute, that will be what $1 / 10$ is that?

JM **Joseph Mugambi** 1:53:43

Because if I convert it per minute, there'll be 360 entities coming in in one minute.

Yet we are serving an entity per minute, which which the way I interpret it is that the system will be clogged.

There's no way the system will be able to serve 360 entities in a minute, yet it's it's taking one minute to serve one entity.

CO **Collins Odhiambo** 1:54:07

No.

OK, I get you. I get, I get, I get. I get the way you're looking at it. Not really, because it's a theoretical approach.

The the idea is that we want there to be a queue. OK, so you want there to be a queue, so.

The relationship between.

Uh, the relationship between.

The way at which entities arrive versus the way at which service is done, what we just make sure for this model to work so that there is a queue is that the the duration it takes to serve a client is more compared to the.

The rate at which customers arrive. That's the only thing. Now the other bit of whether the system is infinite. So those are the assumptions we make with MM1. We

make that the system is infinite. In other words, it can hold infinitely many people. And then the population where they are coming from is also infinite. OK, so if you remember the Kendall way of classification of queues, so this MM one assumes that one the system has the capacity, the population.

Has a infinite capacity. OK, if you if you put put an embargo on the number, the maximum it can hold, then that that will not work in this model.

That means we will have to include some components where the population we can maximize, we can restrict the maximum population. So for example, like in a bank, I don't know, can they stop you until unless there's it reaches 4:00? Is it 4? Let's say if we are closing at 4:00.

They'll only treat people who are who are in anybody after that, they will not treat them or something like that.

OK. So I understand what you're saying, but this, this one I'm just trying to, I was just trying to figure out theoretically so that we think in terms of queue and models, modeling queuing system.

As for the if the system will be clocked, that's another thing in reality if you're if you're modelling because you need to adjust, probably the duration is different, we adjust the duration and things like that. So I but I understand what you're saying.

I just thought this is just an example I came up with to try to fit in into the model and and just have the hypothetical numbers to work in. But reality is that when you start modelling you have to adjust some of those concepts.

Like either restrict even if the server is working this way, then can you restrict it in a way?

I see, especially after COVID-19, most systems become clogged, but I think it all depends on what you're modelling. But there there are all those provisions where you can adjust and allow certain.

Parameters within the like restrict the maximum or make the server work faster. The the only thing is that you just make sure that the rate at which service is done is lower than the rate at which clients come in.



Joseph Mugambi 1:58:08

OK. Thank you.



Collins Odhiambo 1:58:08

But those are very good questions. Those are very good questions that can help

someone to figure out what best you can do. Let me see.

If I can just do a recap again and then.

Next week we will start on birth and death processes.

We'll do birth and death processes.

Next week and then we also do, I think we are meeting twice next week.

From the.

From the timetable that we were given.

From the timetable that we were given, I think we are meeting on Wednesday and Thursday.

Let me look at my program on Wednesday next week.

OK, so we're meeting.

Wednesday and Thursday next week.

Wow, I haven't looking, but that's fine. We can still meet, commit for 1 1/2 or so hours next weekend on Wednesday, then Friday we finish and Thursday we finish up.

But let me just give a summary of our queuing system. So next week we start on.

The path processes, then after that we look at the branching processes and then I think we have martingales, a bit of martingales, theory of martingales.

So next week and the other week we are meeting Wednesday and Thursday and then now we finish up the other two weeks we'll be done.

OK. So just a recap of queuing systems, which we've covered for quite a while now.

The three components of a queuing system is that there has to be arrival, there has to be queue, there has to be service mechanism.

The queuing model theory is just a collection of mathematical model of various queuing systems that takes in input parameters, so they are of those elements and then provide quantitative parameters describing the system performance.

And so because the random nature of the process is involved, the killing theory demands is demanding and all what are based on strong assumptions. So we have a lot of strong assumptions. So like that question you asked me, that's a very strong assumption we were making.

But in reality you have to adjust. I find if the more you adjust, the more complicated it becomes. The more you adjust the assumptions, the more complicated it becomes.

But but that's the nature of the queuing model. Queuing systems are practically very important because of the typical trade off for various cost providing services and cost associated with waiting the service. OK in customer care in.

Marketing, customer service in different areas of services where you require somebody queuing for a service where the queuing system plays a very critical role and it's a stochastic process.

Because we are looking at the state of the system after every different time and within different period of time we have, we have entities coming in and we have entities going out of the system.

Some of the critical things that we have looked at include the fraction of time the survey side or this is more important for decision making. If you're a manager, you manage a branch, you'd want to know how many, what's the fraction of time you are. You are your workers where employees are idle. They are not working on them. They're idle for you to make some decisions.

OK. And or rather you to adjust how you strategically implement your services?

In the in in situations where service is very limited in many aspects here in the United States, I think things like medical services.

A very um.

And I say demand, demanding or anything. Anybody with a medical degree is very marketable because there are few experts who given the strict, given the strict systems that they are put in place before you qualify.

Yeah, to become a specialist, a physician.

I think it should take you like 12 years or so from the time you're starting your first step. And so because of those stigent measures they are put, it's a very, it's a very marketable, demanding. The other day I was, I was booking.

For a check up for my daughter and I was told the only available available appointment is December, end of December. So that's like four months because those specialists are not very rare and they are not there.

So then.

Queuing system become very critical in this kind of an environment because of minimum resources in terms of the expertise you need for that kind of service.

So it becomes very important to model Q. So organisations or individuals who have very limited resources and don't want to schedule services. They they take up, they take a lot of interest in.

How you model your queue, how you manage your queue so that you make very good decisions and more efficient decisions.

What to expect? What is the expected number of customers present in the queue?

We will look at that expected time.

So many here.

So expected time that a customer spends in the queue, that's also very important.

The probability distribution of the number of customers present, we've looked at that. The distribution of the customers waiting time, we've looked at that.

And some of the assumptions we've made is that when you model IQ, you assume the arrival follows Poisson, the distribution of services follows exponential and we have to be clear on the number of servers that are available and the type of.

Um queue what we call the line discipline, which is the queuing the queuing display. Typically we've assumed first in, first out. In some scenarios you can find different types.

The candle classification system. So this one we checked last time and it's just a way of classifying Q and so we focused on the MM one, which is a derivative of the candles.

So this MM one is where we put all of our focus in our model. These are the assumptions for the MM one that the arrival follows Poisson. The population where the arrival originates is infinite, so there is no maximum. And then the question are you asking about the length? There's no limit on the length.

So it can't be clogged. There's no limit. We assume there's no limit. We also assume there's no bulking from the queue. That means once you join the queue, you have to be served. You cannot come out of the queue before you're served. And then the queuing mechanism is before first in, first out.

And one object at a time arrives during at any point of time of intervals. And then of course the service time follows exponential distribution.

So we went ahead and modeled this structure, this, this structure.

Came up with the average waiting time in the system, then came up with the average waiting time in the queue, and then using the Little's law we were able to show the system waiting time is this.

And then the system.

How much Kiwi time is this?

That's just an example and I do think we should stop there for the queuing system so that next week we start off with.

bath and death processes.

Uh, any question?

Like us to stop there for today. So next week we move on to birth and death processes.

Alright, so you enjoy the rest of your evening.
See you next week.

SK **Stephen Korir** 2:08:50

Thank you. Bye.

RJ **Rutendo Julia** 2:35:01

I don't know. All I know is Wendy has already.

● stopped transcription